

Data Mining – Lecture 5

1-Frequent pattern is pattern that occurs frequently in a data set .

- a- True
- b- False

Answer : a.

2-Items that are frequently associated together can be represented as an association rules.

- a- True
- b- False

Answer : a.

3-_____ is a measure of rule interestingness.

- a- Support
- b- Confidence
- c- Both
- d- None

Answer : c.

Computer → antivirus_SW [Support = 2% , Confidence =60%]

“ The upcoming questions are related to this forumal “

4- What does a support value of 2% signify in the given association rule?

- a- 2% of customers buy antivirus_SW
- b- 2% of transactions include both computers and antivirus_SW
- c- 2% of transactions include only computers
- d- 2% of customers buy computers

Answer : b.

5- In the given association rule, what does a confidence value of 60% indicate?

- a- 60% of customers buy computers
- b- 60% of transactions include both computers and antivirus_SW
- c- 60% of transactions include only computers
- d- 60% of customers who buy computers also buy antivirus_SW

Answer : d.

6- Which measure of rule interestingness indicates the percentage of transactions that contain both items?

- a- Support
- b- Confidence
- c- Lift
- d- Conviction

Answer : a.

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7- If 1000 transactions were analyzed, how many transactions would be expected to include both computers and antivirus_SW based on the support value?

- a- 20
- b- 50
- c- 40
- d- 100

Answer : a.

8- If 80 customers purchased computers, how many of them are expected to have also purchased antivirus_SW based on the confidence value?

- a- 48
- b- 32
- c- 60
- d- 20

Answer : a.

9- What does a confidence value of 100% imply in the context of the given association rule?

- a- Every customer buys computers.
- b- Every customer buys antivirus_SW.
- c- Every transaction includes both computers and antivirus_SW.
- d- Every customer who buys computers also buys antivirus_SW.

Answer : d.

Ex: Let $min_Sup. = 50\%$, $min_conf. = 50\%$

Frequent Patterns:

$\{A:3, B:3, D:4, E:3, AD:3\}$

Transaction-id	Items bought
10	A, B, D
20	A, C, D
30	A, D, E
40	B, E, F
50	B, C, D, E, F

10- What is the confidence of the rule $D \rightarrow A$?

- a- 60%
- b- 75%
- c- 100%
- d- 80%

Answer : b.

11-- What is the confidence of the rule $A \rightarrow D$?

- a- 60%
- b- 75%
- c- 100%
- d- 80%

Answer : c.

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12- $A \rightarrow D$ is always implies $D \rightarrow A$.

- a- True
- b- False

Answer : b.

13- Apriori property implies that all nonempty subsets of a frequent itemset ____ also be frequent.

- a- Must
- b- may
- c- may not
- d- none

Answer : a.

14- If any subset is not frequent , the superset must be ____.

- a- Frequent
- b- Infrequent
- c- Both
- d- None

Answer : b.

15- If $\{A,B\}$ is a frequent superset then ____ .

- a- $\{A\}$ must be frequent
- b- $\{B\}$ must be frequent
- c- $\{A,B\}$ must be frequent
- d- A and B

Answer : d, explanation (“ based on the Apriori property , the subsets must be frequent if the superset is)

16- What is the support count for item I2?

- a- 6
- b- 7
- c- 8
- d- 9

Answer : c.

17- What is the support count for the itemset $\{I1, I2, I3\}$?

- a- 2
- b- 5
- c- 6
- d- 7

Answer : a.

Transactional data example
N=10, min_supp count=2

TID	List of items
T100	I1, I2, I5
T200	I2, I4
T300	I2, I3
T400	I1, I2, I4
T500	I1, I3
T600	I2, I3
T700	I1, I3
T800	I1, I2, I3, I5
T900	I1, I2, I3
T1000	I1, I2

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Before we get into Mining frequent patterns (FP-Growth) , if you couldn't understand well, watch this video (Arabic explanation) : <https://www.youtube.com/watch?v=0Mlj3Dg9BBU>

18- Which item has the highest support count or frequency :

- a- I1
- b- I2
- c- I3
- d- I4

Answer : b .

“To be honest those kind of questions seems quite useless , I think such Questions must be written with atleast 10 marks , because common , How could we focus on an MCQ related to 10-20 minutes of work ? “

19- When does a lift value less than 1 indicate in terms of association between items A and B?

- a- A and B are independent
- b- A is negatively correlated with B
- c- A is positively correlated with B
- d- A and B are equally correlated

Answer : b.

20- If the lift value between two items A and B is 1.5, what can be inferred about their association?

- a- A is negatively correlated with B
- b- A is positively correlated with B
- c- A and B are independent
- d- A and B are equally correlated

Answer : b.

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T400	I1, I2, I4
T500	I1, I3
T600	I2, I3
T700	I1, I3
T800	I1, I2, I3, I5
T900	I1, I2, I3
T1000	I1, I2

" شكرا لكم ولا تنسونا بالدعاء "